1-3-c. Refraction of light from optical interfaces:

In Fig.1-16. when light is incident at an interface-the ray is reflected at the interface surface according to the law of reflection and refracted into the second medium. The bending of the light ray at the interface is called refraction.

1-Index of refraction[9,21] :

Def.: Interface

The interface surface is a transparent surface separate between two transparent optical media (air and glass or glass and water).

The two transparent optical media that form an interface are distinguished from one another by a constant called the index of refraction (n) see table 1-1.What is the relation between index of refraction and the Hardnesss[5]?

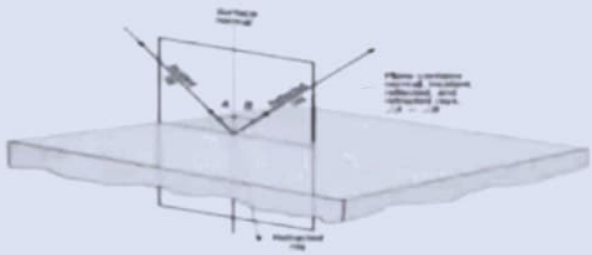


Fig.(1-16) .Reflection and Refraction at an interface.

Media	Index of Refraction
Vacuum	1.00
Air	1.0003
Carbon dioxide gas	1.0005
Ice	1.31
Pure water	1.33
Ethyl alcohol	1.36
Quartz	1.46
Vegetable oil	1.47
Olive oil	1.48
Acrylic	1.49
Table salt	1.51
Glass	1.52
Sapphire	1.77
Zircon	1.92
Cubic zirconia	2.16
Diamond	2.42
Gallium phosphide	3.50

n & Hardness

Table 1-1. Index of Refraction for **various Materials** at 589 nm.

meet.google.com is sharing your screen. Stop sharing Hide

تم مشاركة العرض من قبل wageda.

The index of refraction (n):(What Happens to the angle of refraction?)

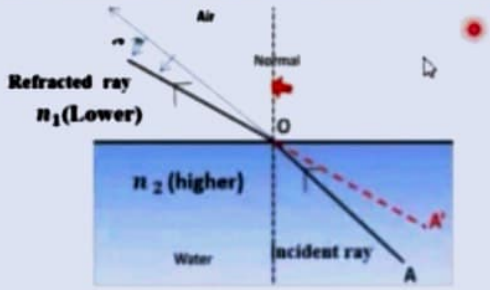
where C=Speed of light in free space , V=Speed of light in the medium and n in free space =1

$$n=\frac{C}{V} \qquad \{8\text{-}1\}$$

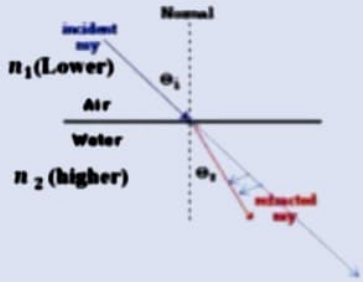
***From Fig.(1-17- a and b) ,**

***in the first case a (lower-to higher refractive index)the light ray is bent toward the normal. (a)**

*** in the first case b (higher-to lower refractive index)the light ray is bent away from the normal. (b)**



b) Higher to lower bending away from normal.



a) Lower to higher bending toward normal.

Fig. (1-17- a and b)

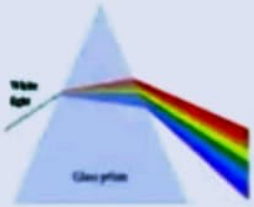
•As light travels from one medium to another, its frequency (f) does not change but its wavelength (λ) does.

$$f_1 = f_2 = f = 1/t \quad \{9\text{-}1\}$$

$$V = \lambda / t \quad \{10\text{-}1\}$$

$$V = \frac{distance}{time} = \frac{d}{t} = d\,(1/t) = \lambda f \quad \{11\text{-}1\}$$

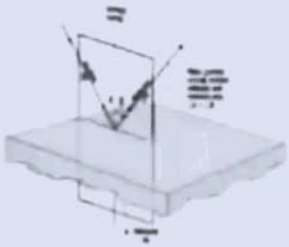
$$\therefore V_1 = f\lambda_1 \quad , V_2 = f\lambda_2 . \quad \{12\text{-}1\}$$



Color	λ (nm)	Freq (Hz)
Red	760-647	4.3×10^{14}
Orange	647-585	4.3×10^{14}
Yellow	585-575	5.2×10^{14}
Green	575-491	5.6×10^{14}
Blue	491-424	6.6×10^{14}
Violet	424-380	7.3×10^{14}

From equation { 8-1 } where $n = c/v$, $n1 = c/v1$
$n2 = c/v2$

$$\therefore \frac{V_1}{V_2} = \frac{c/n_1}{c/n_2} = \frac{f\lambda_1}{f\lambda_2} \quad \therefore \frac{c/n_1}{c/n_2} = \frac{\lambda_1}{\lambda_2} \quad \{13\text{-}1\}$$



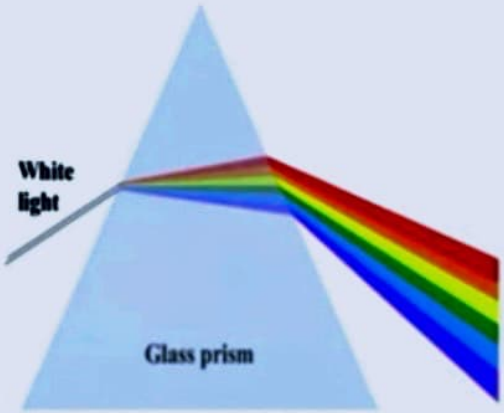
$$\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} \quad \therefore n_1\lambda_1 = n_2\lambda_2 , \quad \ldots n_1 = 1 \ldots \text{in vacuum.} \quad \{14\text{-}1\}$$

$$\therefore \lambda_1 = n_2\lambda_2 \quad \{15\text{-}1\}$$

$$\therefore n = \frac{\lambda}{\lambda_n} \quad \ldots\ldots\ldots\ldots \; n > 1 , \; \lambda_n < \lambda . \quad \{16\text{-}1\}$$

meet.google.com is sharing your screen. Stop sharing Hide

{ 10-1}
{ 11-1}
{ 12-1}



Color	λ (nm)	Freq (Hz)
Red	760-647	4.3×10^{14}
Orange.	647-585	4.3×10^{14}
Yellow	585-575	5.2×10^{14}
Green	575-491	5.6×10^{14}
Blue	491-424	6.6×10^{14}
Violet	424-380	7.3×10^{14}

c/v1
c/v2

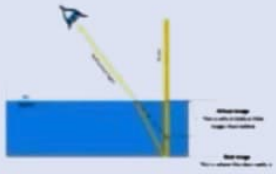
{ 13-1}

meet.google.com is sharing your screen. Stop sharing Hide

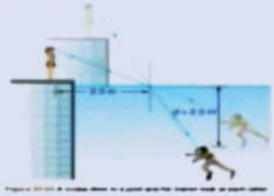
3- The application on refraction: as shown in Fig. (1-17- c, d, e, f, g, h and i)

C)

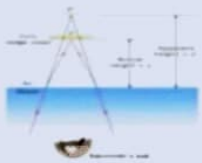
d)



e)



F)



The mirage[23]

g)

h)

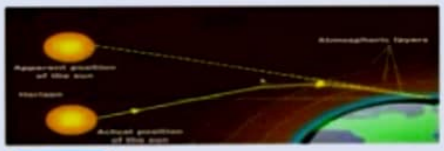
i)

Fig. (1-17- c, d, e, f, g, h and i)

Question:

How can we see the sun after disappearing back to the horizon ?

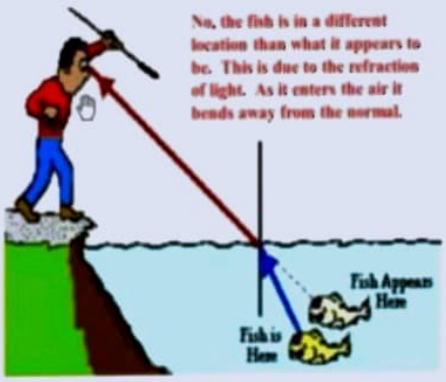
Answer: According to the law of refraction



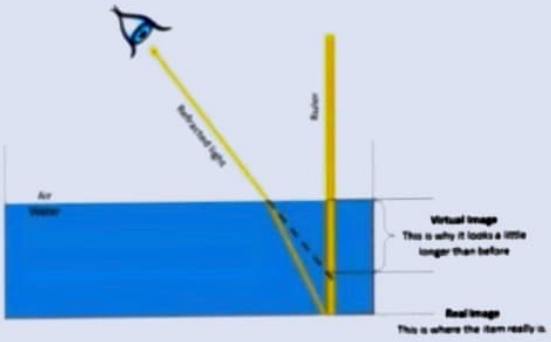
meet.google.com is sharing your screen. Stop sharing Hide

3- The application on refractio

C)



d)



e)

The mirage[23]

shown in Fig. (1-17- c, d, e, f, g, h and i)

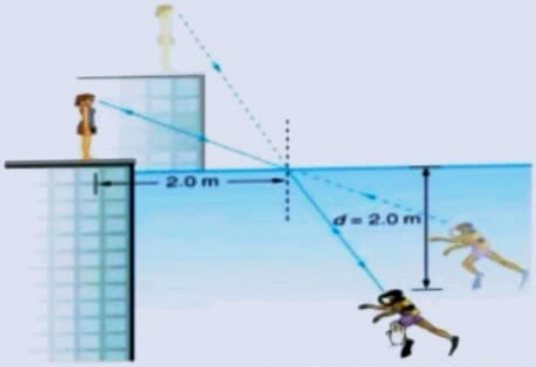
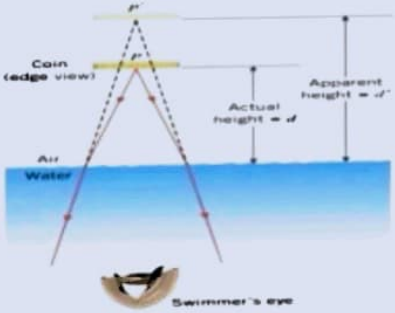


Figure 25-53 A scuba diver in a pool and his trainer look at each other

F)



1-5-c. Refraction of light from optical interfaces:

In Fig.10. when light is incident at an interface-the ray is reflected at the interface surface according to the law of reflection, and refracted into the second medium. The bending of the light ray at the interface is called refraction.

1-Index of refraction:

Def.: Interface

The interface surface is a transparent surface separate between two transparent optical media(air and glass or glass and water).

The two transparent optical media that form an interface are distinguished from one another by a constant called the index of refraction (n).

n in free space =1

$$n=\frac{C}{V}$$

C=Speed of light in free space

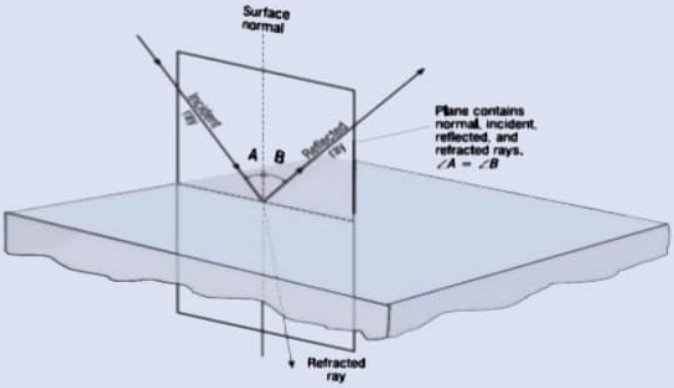


Fig.(1- 10) .Reflection and Refraction at an interface.

Media	Index of Refraction
Vacuum	1.00
Air	1.0003
Carbon dioxide gas	1.0005
Ice	1.31
Pure water	1.33
Ethyl alcohol	1.36
Quartz	1.46
Vegetable oil	1.47
Olive oil	1.48
Acrylic	1.49
Table salt	1.51
Glass	1.52
Sapphire	1.77
Zircon	1.92
Cubic zirconia	2.16
Diamond	2.42
Gallium phosphide	3.50

n & Hardness

Table 1-1. Index of Refraction for various Materials at 589 nm.

*From Fig.(1- 11 a and b) ,

*in the first case a (lower-to higher refractive index)the light ray is bent toward the normal. (a)

* in the first case b (higher-to lower refractive index)the light ray is bent away from the normal. (b)

2 – The relation between the refractive index and wavelength

*As light travels from one medium to another, its frequency does not change but its wavelength does.

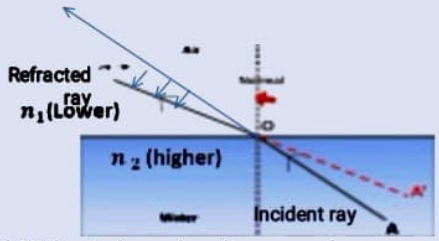
$$f_1 = f_2 = f$$

$$V = \frac{distance}{time} = \frac{d}{t} = \lambda f, \qquad \therefore V_1 = f\lambda_1 \quad , V_2 = f\lambda_2 \qquad \therefore \frac{V_1}{V_2} = \frac{C/n_1}{C/n_2}\,\frac{f\lambda_1}{f\lambda_2} \quad \therefore \frac{C/n_1}{C/n_2} = \frac{\lambda_1}{\lambda_2}$$

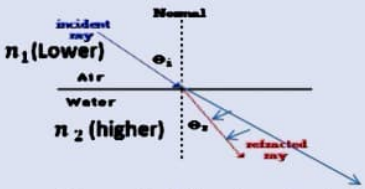
$$\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} \quad \therefore n_1\lambda_1 = n_2\lambda_2 \text{ , ... } n_1 = 1 \text{ ... in vacuum.}$$

$$\therefore \lambda_1 = n_2\lambda_2$$

$$\therefore n = \frac{\lambda}{\lambda_n} \qquad \text{.....} \quad n > 1 \text{ , } \lambda_n < \lambda \quad .$$



b) Higher to lower bending away from normal.



a) Lower to higher bending toward normal.

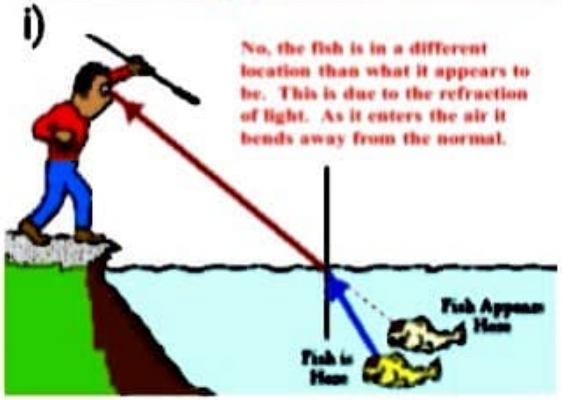
Fig. (1-11 a and b)

b) Higher to lower bending away from normal.

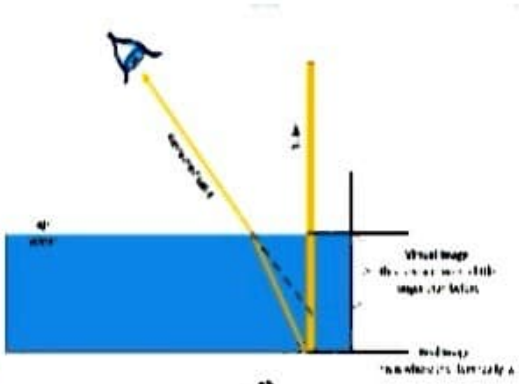
a) Lower to higher bending toward normal.

3- The application on refraction: as shown in Fig. (1-11 c, d, e, f, g, h and

i)



c)



d)

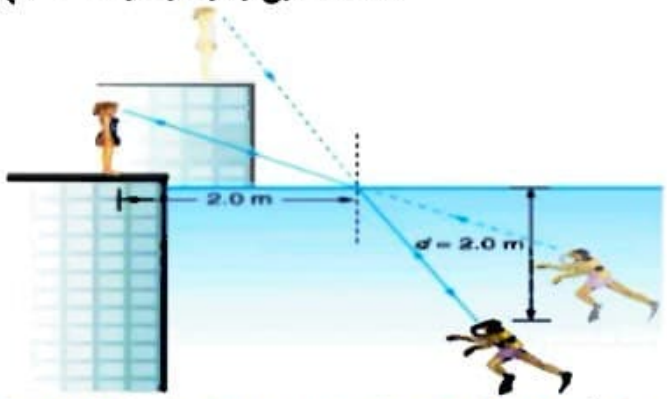
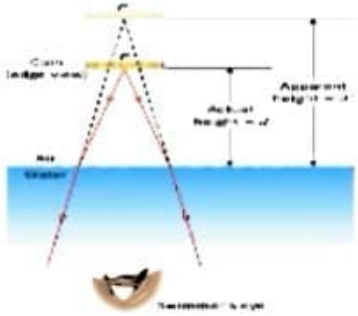


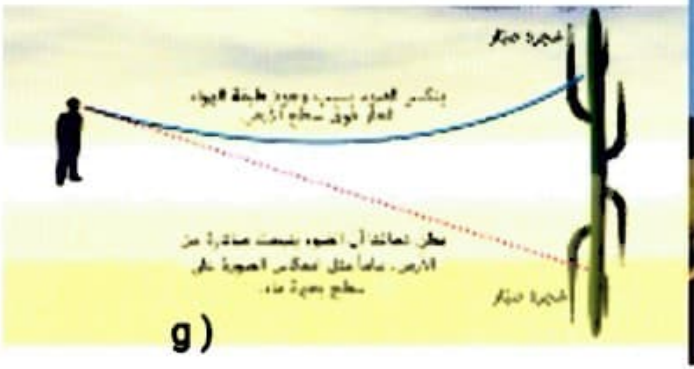
Figure 25.53 A scuba diver in a pool and his trainer look at each other

e)



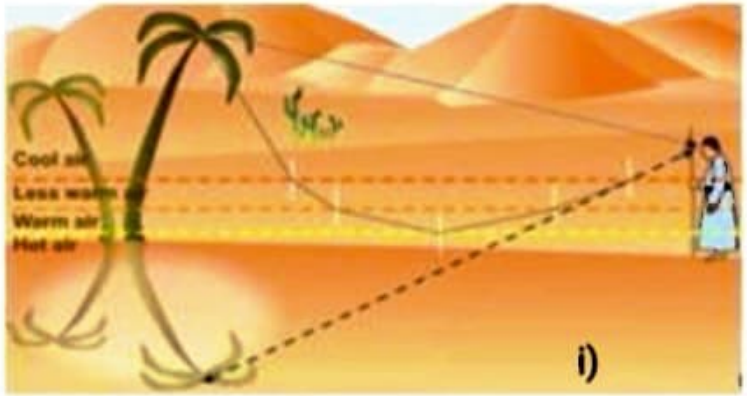
f)

The mirage



g)

h)

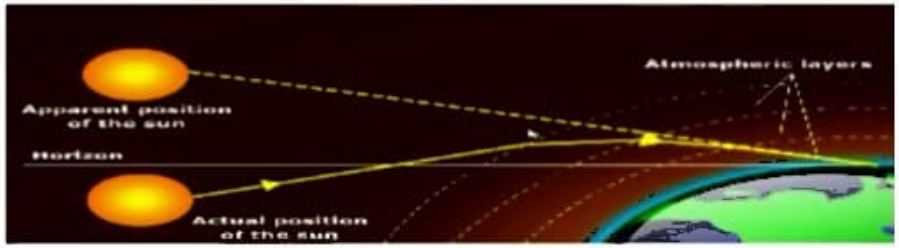


i)

Question:

How can we see the sun after disappearing back to the horizon ?

Answer: According to the law of refraction



1-4-b Critical angle and total internal reflection[24]:

When light travels from a medium of higher refractive index to one of lower refractive index ,we encounter some interesting result, which called total internal reflection ,as shown in Fig.(1-19).

Comment:

1- The first ray(1) is normal in the first medium with greater refractive index (n_i) ,it suffers any change in the second medium with lower refractive index (n_r).(?)

2-The second and the third ray(2,3) bending away from the normal.

3- The fourth ray(4), the incident with a critical angle ic , causes the refracted ray to bend by 90°.

4-the fifth ray(5),is incident on interface at an angle greater than the critical angle (ic).Ray (5) is totally reflected into the same medium. By using Snell's law ,then we prove the following:

$$n_1 sin\theta_1 = n_2 sin\theta_2 \therefore n_i sin\theta_c = n_r sin90^0 \therefore \; i_c = sin^{-1}\left(\frac{n_r}{n_i}\right)$$

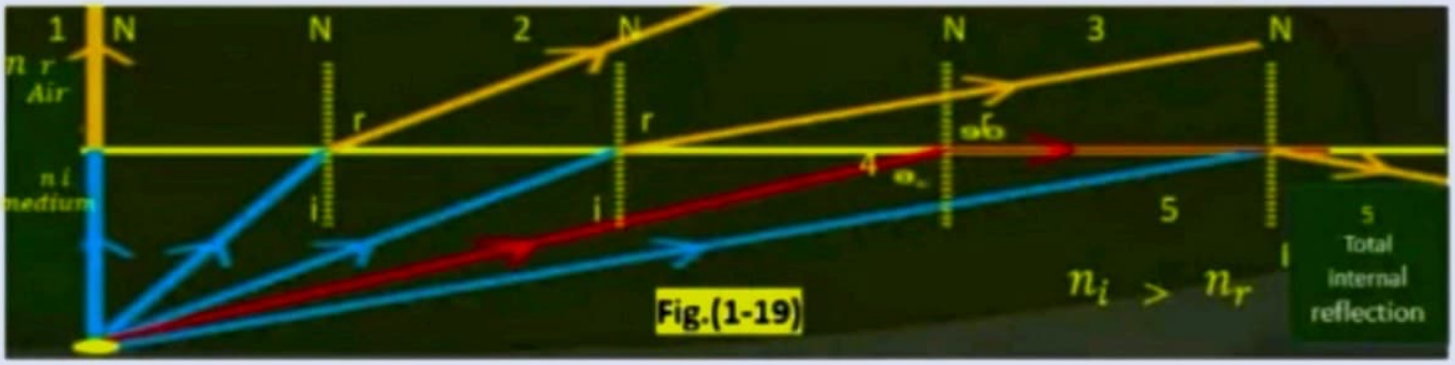


Fig.(1-19)

meet.google.com is sharing your screen. Stop sharing Hide

1-4-a- Snell's law[22] .

The mathematical expression of Snell's law and an accompanying drawing are given in Fig.(1-18).

Fig.(1-18).

$$\sin\theta_1 = \frac{BB'}{AB'} \qquad \sin\theta_2 = \frac{AA'}{AB'}$$

Dividing side by side $\frac{\sin\theta_1}{\sin\theta_2} = \frac{BB'}{AA'}$

or

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{V_1 \Delta t}{V_2 \Delta t} \quad \text{or} \quad \boxed{\frac{\sin\theta_1}{\sin\theta_2} = \frac{V_1}{V_2}} \quad (1)$$

Since $n = \frac{c}{V}$; therefore, $n_1 = \frac{c}{V_1}$ and $n_2 = \frac{c}{V_2}$

Equation (1) becomes $\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1}$ and $\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}$ Snell's Law

meet.google.com is sharing your screen. Stop sharing Hide

1-5-Hygens's principle[26] applied to Reflection and Refraction: Fig.(1-22- a, b, c, d)

a) b) c) d)

Fig.(1-22- a, b, c, d)

Huygens's principle depends on the Snell's wavelets (in his refraction law) and the idea of the congruent of the two triangles.

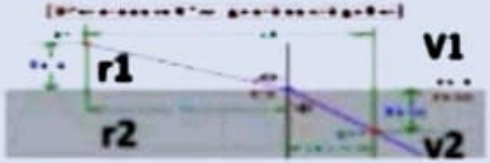
Reflection law: $\theta_1 = \theta_1{}'$ ... ...**Refraction law:** $n_1 sin\theta_1 = n_2 sin\theta_2$

1-6-Fermat's Principal[27] as shown in Fig.(1-23- a, b)

Fermat's Principal :When a light ray travels between any two points P and P/ , its actual path will be the one that requires the least time.

Reflection law: $\theta = \emptyset$ .....**Refraction law:** $n_1 \sin\theta = n_2 \sin\emptyset$

a)



b)

Fig.(1-23- a, b)

1-6- Snell's law.

The mathematical expression of snell's law and an accompanying drawing are given in Fig.(1-12).

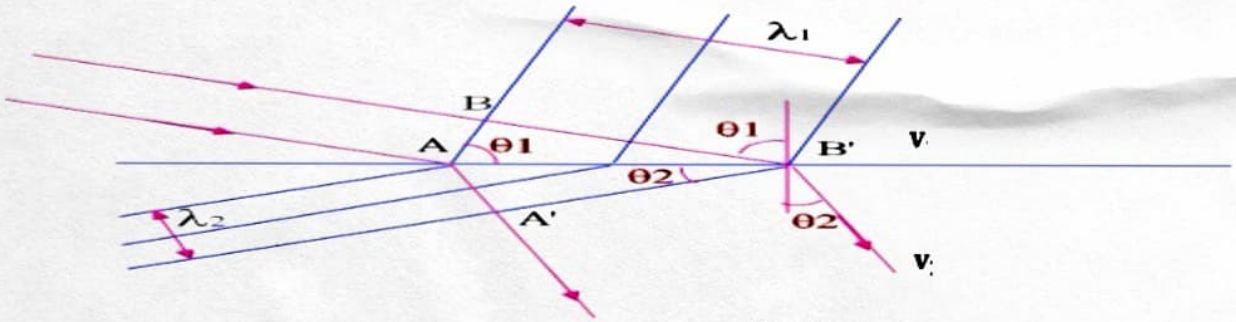


Fig.(1-12).

$$\sin\theta_1 = \frac{BB'}{AB'}$$

$$\sin\theta_2 = \frac{AA'}{AB'}$$

Dividing side by side : $\frac{\sin\theta_1}{\sin\theta_2} = \frac{BB'}{AA'}$

or

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{V_1\,\Delta t}{V_2\,\Delta t} \quad \text{or} \quad \boxed{\frac{\sin\theta_1}{\sin\theta_2} = \frac{V_1}{V_2}} \quad (1)$$

Since $n = \frac{c}{V}$; therefore, $n_1 = \frac{c}{V_1}$ and $n_2 = \frac{c}{V_2}$

Equation (1) becomes: $\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1}$ and $\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}$

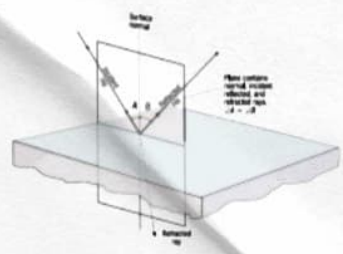
Snell's Law

Example 1:

The light pass from the air to the glass with refractive index = relative refractive index= 1.49 .the incident angle is (52^0) what is the angle of refraction.

Answer:

$$n_1 sin\theta_1 = n_2 sin\theta_2$$



$n = \frac{n_2}{n_1} = \frac{sin\theta_1}{sin\theta_2}$

$n = \frac{sin\theta_1}{sin\theta_2}$

$sin\theta_2 = \frac{sin\theta_1}{n} = \frac{\sin 52^0}{1.49} = \frac{0.78}{1.49}$

$\theta_2 = sin^{-1} x\ 0.52$

$\theta_2 = 31.33^0$

Example 2:

The incident ray on one face of rectangular glass is (50^0)and its refractive index is (1.5).When Part of the incident ray is reflected and the other part is refractive ,what is the angle between the reflected and the refracted rays.

Answer:

$sin\theta_2 = \frac{sin\theta_1}{n} = \frac{\sin 50^0}{1.5} = \frac{0.76}{1.5} = 0.51$

$\theta_2 = sin^{-1} x\ 0.51$

$\theta_2 = 30.7$

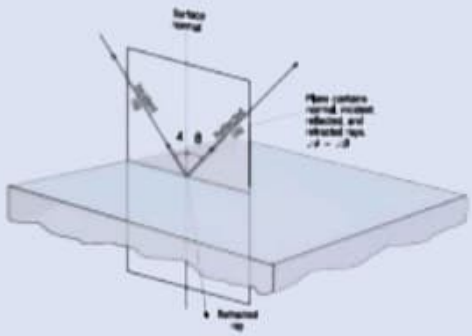
$\theta = (90-30.7) + (90-50) = 59.3+40 = 99.3^0$

Example 1:

The light pass from the air to the glass with refractive index = relative refractive index= 1.49 .the incident angle is (52^0) what is the angle of refraction.

Answer:

$$n_1 sin\theta_1 = n_2 sin\theta_2$$



$n = \frac{n_2}{n_1} = \frac{sin\theta_1}{sin\theta_2}$

$n = \frac{sin\theta_1}{sin\theta_2}$

$sin\theta_2 = \frac{sin\theta_1}{n} = \frac{\sin 52^0}{1.49} = \frac{0.78}{1.49}$

$\theta_2 = sin^{-1} \times 0.52$

$\theta_2 = 31.33^0$

Example 2:

The incident ray on one face of rectangular glass is (50^0)and its refractive index is (1.5).When Part of the incident ray is reflected and the other part is refractive ,what is the angle between the reflected and the refracted rays.

Answer:

$sin\theta_2 = \frac{sin\theta_1}{n} = \frac{\sin 50^0}{1.5} = \frac{0.76}{1.5} = 0.51$

$\theta_2 = sin^{-1} \times 0.51$

$\theta_2 = 30.7$

$\theta = (90-30.7) + (90-50) = 59.3+40 = 99.3^0$

1-6-b Critical angle and total internal reflection:

When light travels from a medium of higher index to one of lower index ,we encounter some interesting result, as shown in Fig.(1-13).

Comment:

1- The first ray(1) is normal in the first medium with greater refractive index (n_i) ,it suffers any change in the second medium with lower refractive index (n_r).(?)

2-The second and the third ray(2,3) bending away from the normal.

3- The fourth ray(4), the incident with a critical angle ic , causes the refracted ray to bend by 90°.

4-the fifth ray(5),is incident on interface at an angle greater than the critical angle (ic).Ray (5) is totally reflected into the same medium. By using Snell's law ,then we prove the following:

$$n_1 sin\theta_1 = n_2 sin\theta_2 \therefore n_i sin\theta_c = n_r sin90^0 \therefore \; i_C = sin^{-1}\left(\frac{n_r}{n_i}\right)$$

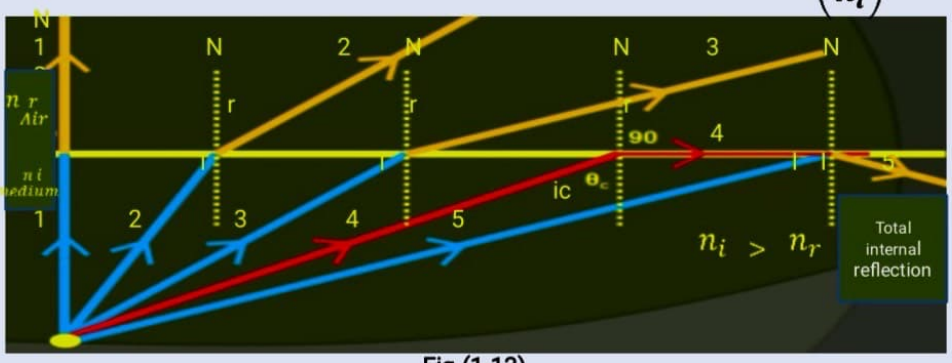


Fig.(1-13)

Example3:

What is the critical angle for the total internal reflection, when the ray of light transmitted from glass with refractive index 1.52 to water with refractive index 1.33.

Answer:

Sin $\emptyset_c = \frac{n_2}{n_1}$

Sin $\emptyset_c = \frac{1.33}{1.52} =$ =, =$\emptyset_c$ = 61.0°

Example 4:

Find the critical angle for snow , with refractive index 1.31 in the very cold regions. What is the differences between the optical fiber (Snow optical fiber)which manufactured from snow and that from glass (glass optical fiber).

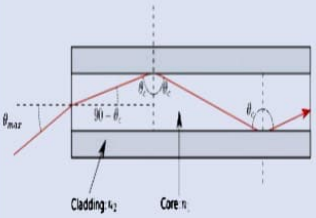
Answer:

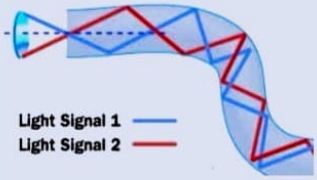
$\emptyset_c = sin^{-1}\left(\frac{n_{air}}{n_{ice}}\right) = sin^{-1}\left(\frac{1.00}{1.31}\right)$

$\emptyset_c$ = 49.8°

The greater the critical angle in snow , the decrease of quantity of the total internal reflection, which lead to decrease to the of quantity of the rays and also the picture will be not clear as it is in the glass optical fiber.

1-6-c-Applications on the Critical angle and total internal reflection of light in industrial production as shown in Fig.(1-14 a , b ,c and d) .





a) Optic fiber.

b) Internet and.. wires.

c)Abe device.

d)Dancing fountain

7-1- Dispersion of light[28]: as shown in Fig (1-24-a, b ,c and d)

Def.: Dispersion is a process, which happens after the refraction light through any surface such as the prism.

Fig.(1-24-a).Shows a normal dispersion curve of n_λ versus λ for different types of optical glass.

Fig.(1-24-b). Shows the separation of individual colors in white light- 400nm-700nm- after passing through a prism.

Fig.(1-24-c).Shows How can we get the white light again after dispersion.

Fig. (1-24-d). Shows the relation between the color , Wavelength and frequency of the diffracted light.

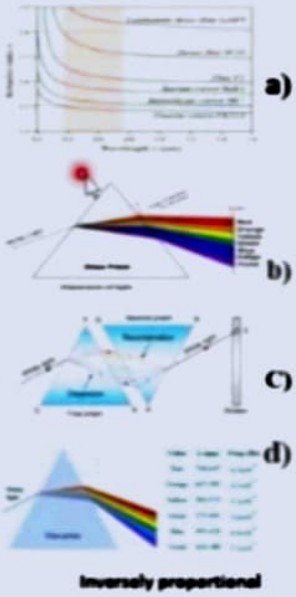
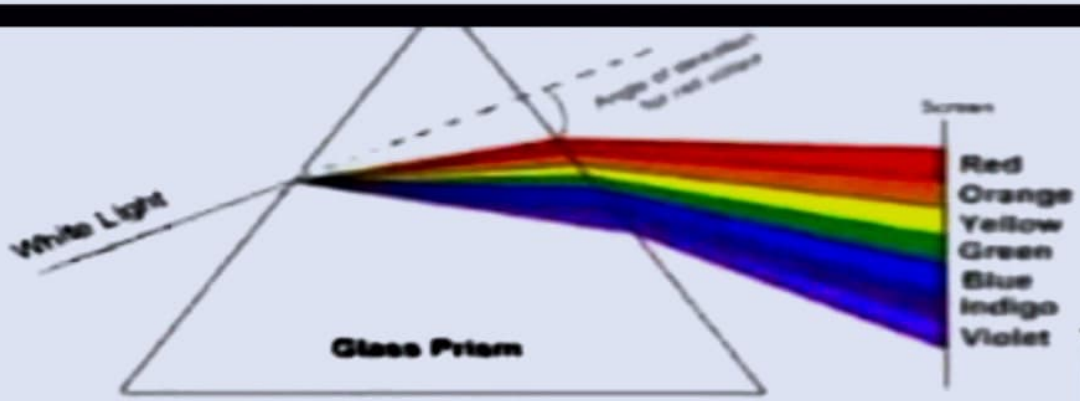
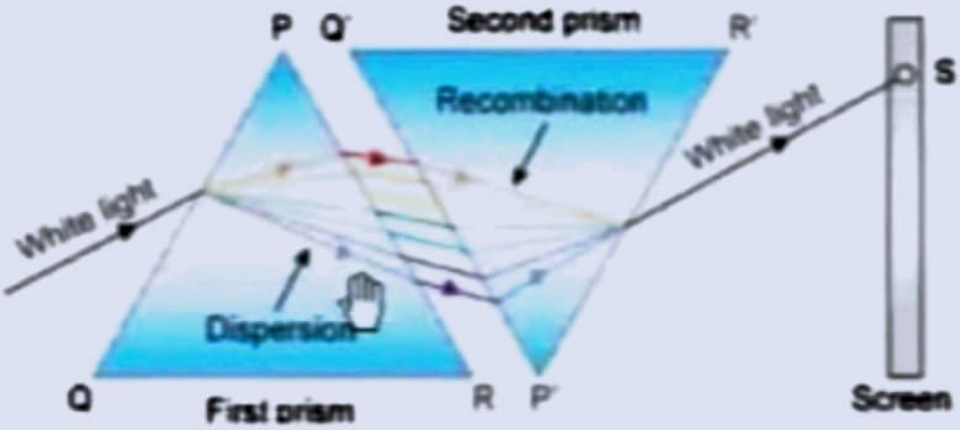


Fig (1-24-a, b ,c and d)



Dispersion of light

b)

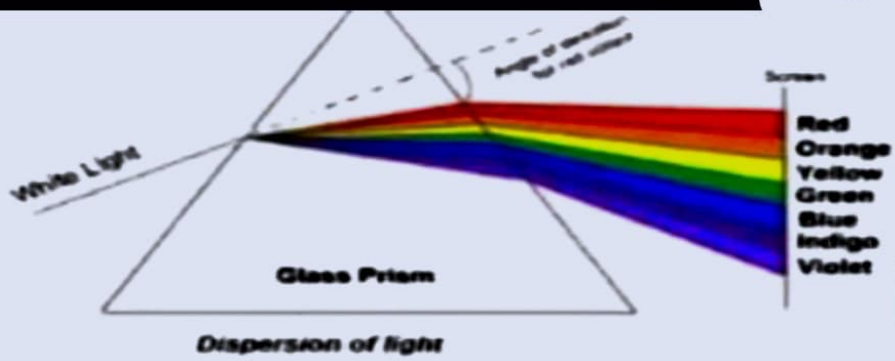


C)

on.

d)

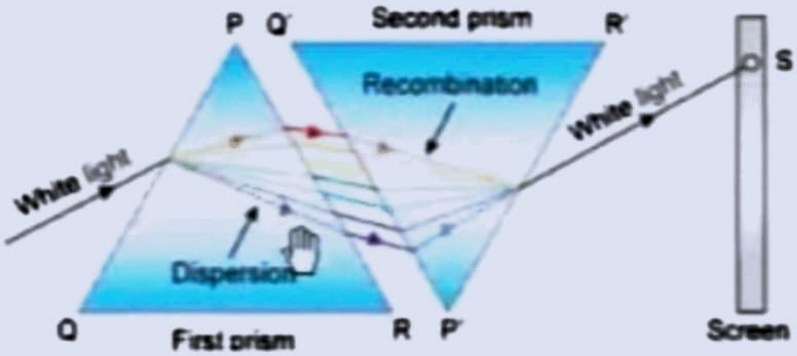
Color	λ (nm)	Freq (Hz)
Red	760-647	4.3×10^{14}



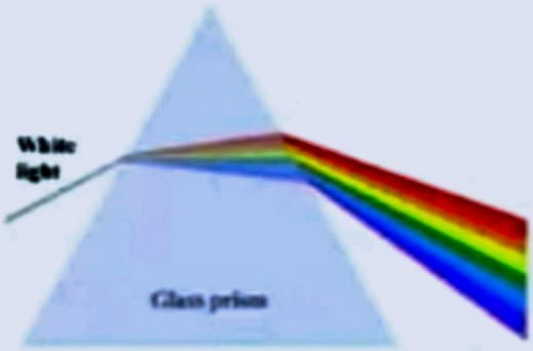
Dispersion of light

b)

on.



C)



Color	λ (nm)	Freq (Hz)
Red	760-647	4.3×10^{14}
Orange	647-585	4.3×10^{14}
Yellow	585-575	5.2×10^{14}
Green	575-491	5.6×10^{14}
Blue	491-424	6.6×10^{14}
Violet	424-380	7.3×10^{14}

d)

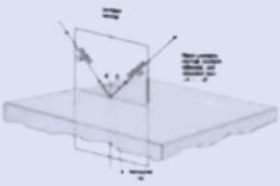
Inversely proportional

Example 1-1:

The light pass from the air to the glass with refractive index = relative refractive index= 1.49 .the incident angle is (52^0) what is the angle of refraction ?.

Answer-1-1:

$$n_1 sin\theta_1 = n_2 sin\theta_2$$



n= $\frac{n_2}{n_1} = \frac{sin\theta_1}{sin\theta_2}$ = relative refractive index

n = $\frac{sin\theta_1}{sin\theta_2}$

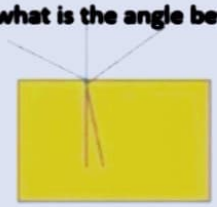
$sin\theta_2 = \frac{sin\theta_1}{n} = \frac{sin\ 52^0}{1.49} = \frac{0.78}{1.49}$

$\theta_2 = sin^{-1}$ x 0.52 ..... θ_2 =31.33^0

Example 1-2:

The incident ray on one face of rectangular glass is (50^0) and its refractive index is (1.5).When Part of the incident ray is reflected and the other part is refractive ,what is the angle between the reflected and the refracted rays ?.

Answer 1-2:



$sin\theta_2 = \frac{sin\theta_1}{n} = \frac{sin\ 50^0}{1.5} = \frac{0.76}{1.5}$ =0. 51

$\theta_2 = sin^{-1}$ x 0.51

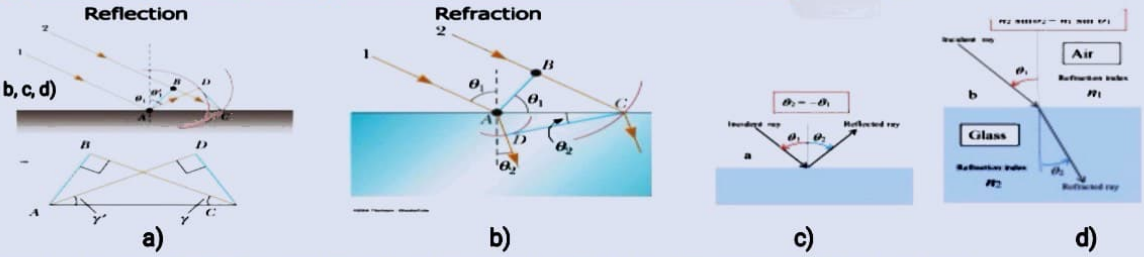
θ_2 = 30.7

θ = (90-30.7) + (90-50) =59.3+40= 99.3^0

meet.google.com is sharing your screen. Stop sharing Hide

1-7-Hygens's principle applied to Reflection and Refra a, b, c, d)

Fig.(1-15 a, b, c, d)



Hygins's principle depend on the Snell's wavelets (in his refraction law) and the idea of the congruent of the two triangle.

Reflection law: $\theta_1 = \theta_1{}'$

Refraction law: $n_1 sin\theta_1 = n_2 sin\theta_2$

1-8. Fermat's Principal: as shown in Fig.(1-16 a, b)

Fermat's Principal :When a light ray travels between any two points P and P\ , its actual path will be the one that requires the least time.

Reflection law: $\theta = \phi$

Refraction law: $n_1 \sin\theta = n_2 \sin\phi$

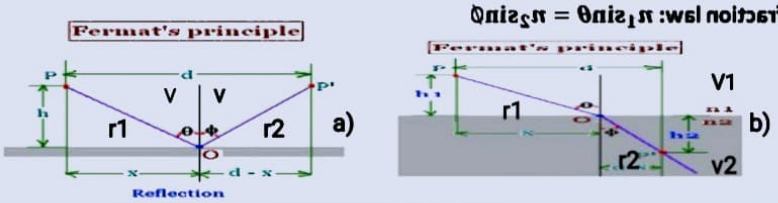


Fig.(1-16 a, b)

1-9. Dispersion of light: as shown in Fig (1-17.a, b ,c and d)

Def.: Dispersion is a process, which happens after the refraction of light through any surface such as the prism.

Fig.(1-17.a). Shows a normal dispersion curve of n_λ versus λ for different types of optical glass.

Fig.(1-17.b). Shows the separation of individual colors in white light- 400nm-700nm- after passing through a prism.

Fig. (1-17.c). Shows How can we get the white light again after dispersion.

Fig. (1-17.d). Shows the relation between the color , Wavelength and frequency of the diffracted light.

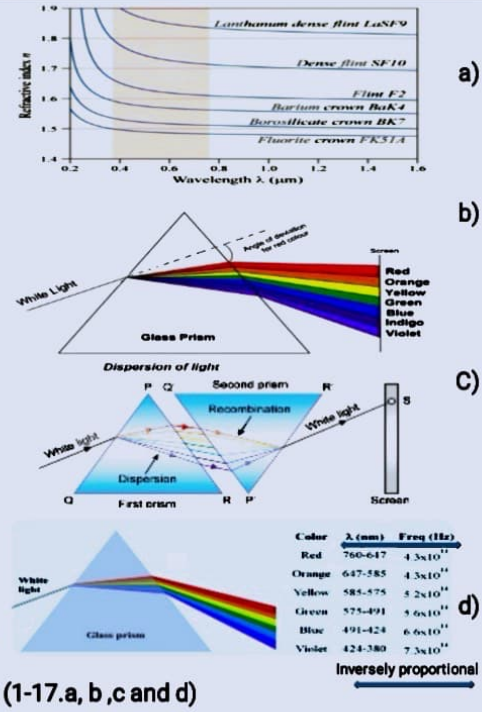


Fig (1-17.a, b ,c and d)

Choose the correct answer from the following ?(c , a, b, b , a)

1-Why the spherical surface mirrors , are capable of forming distinct images?
a) The successive surface tangents along the curved surface are ordered(not random) sections of "plane mirrors ".
b) The surface are smoothly connected.
c) a and b together.

2-Why it is more difficult to see while driving on a rainy night?
a) If the road is wet ,the smooth surface of the water specularly reflects most of your headlight beams away from your eyes.
b) The rough surface diffusely reflects part of your headlight beams back toward you.
c) The interface surface is refracts most of your headlight beams away from your eyes.

3- The Index of refraction of is proportional with:
a) Inversely proportional with speed of light of all wavelength in the medium.
b) Directly proportional with Hardness.
c) a and b together.

4- Why Gallium Arsenide ,germanium and silicon is valuable than diamond?
a) It have a high refractive index.
b)It is very important in all manufacture in the industry.
c) It have law prize.

5-How can we see the sun after disappearing back to the horizon ?
a) According to the law of refraction.
b) According to the law of reflection.
c) According to the law of interference.

Put wright or wrong in front of the following? (/,x ,x ,/, /)

1-The interface surface is a transparent surface separate between two transparent optical media(air and glass or glass and water).
2-Why it is more serious (difficult) to see while driving on a rainy night?
3- We measure Index of refraction for various materials only at wavelength of light equal to 589 nm.
4-As light travels from one medium to another, its frequency does not change but its wavelength does.
5-In free space ,the refractive index (n) is exactly one ,also it is taken for most calculation for gases to be one but for other materials it have a values greater than one.

Example 1-3:

What is the critical angle for the total internal reflection, when the ray of light transmitted from glass with refractive index 1.52 to water with refractive index 1.33.

Answer1-3:

$n_1 sin\theta_1 = n_2 sin\theta_2$

n_1 Sin$\emptyset_c = n_2$ Sin90 Sin $\emptyset_c = \frac{n_2}{n_1}$ …… Sin $\emptyset_c = \frac{1.33}{1.52} =$ =, $= \emptyset_c = 61.0°$

Example 1- 4:

Find the critical angle for snow , with refractive index 1.31 in the very cold regions. What is the differences between the optical fiber (Snow optical fiber)which manufactured from snow and that from glass (glass optical fiber). Were the refractive index of air and water are 1.00 and 1.33 respectively .

Answer1-4:

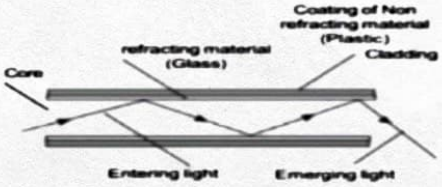
$\emptyset_c = sin^{-1}\left(\frac{n_{air}}{n_{ice}}\right) = sin^{-1}\left(\frac{1.00}{1.31}\right)$…… $\emptyset_c = 49.8°$

Example 1-5:

A pipe of optical fiber , with refractive index 1.53 What is the lower incident angle inside the pipe, which lead to total internal reflection , when the pipe is surrounded by : a-air? . b-water ?. Were the refractive index of air and water are 1.00 and 1.33 respectively .

Answer1-5:

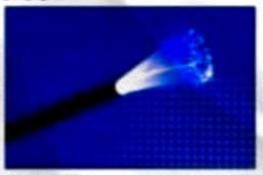
Sin $\emptyset_c = \frac{n_2}{n_1}$ …… $\emptyset_c = sin^{-1}\left(\frac{n_{air}}{n_{glass}}\right) = sin^{-1}\left(\frac{1.00}{1.53}\right) = 40.8°$

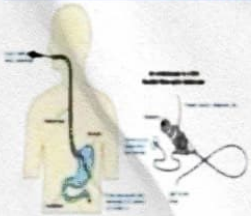


APPLICATIONS OF TOTAL INTERNAL REFLECTION

Fiber optics:

- Telecommunications
  - signals are not affected by electrical storms
  - more signals can be carried with less energy loss
  - the cables are smaller and lighter
- Medicine – Endoscopes
  - One bundle carries light into the required area, while another bundle carries an image of the area back to a monitor.



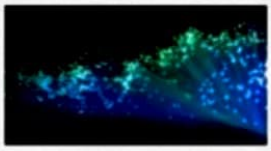


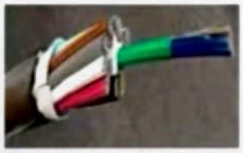
Applications of Total Internal Reflection

- Fibre Optics
  - Fibre optics have revolutionized all forms of communication, including the internet
  - Allows information to be sent as pulses of light instead of pulses of electricity
  - Optical fibres are made of a glass core which is surrounded by an *optical cladding*
  - The cladding is a covering made of a different kind of glass than the fibre inside

meet.google.com is sharing your screen. Stop sharing Hide

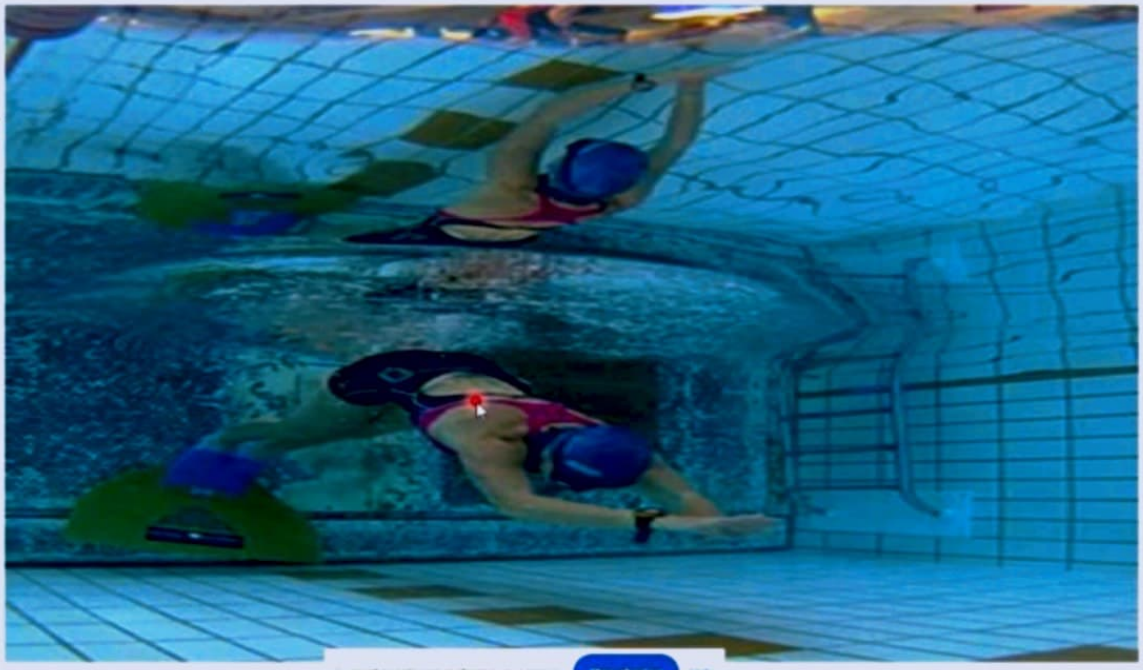
1-4-c-Applications on the Critical angle and total internal reflection of light in industrial production. Fig.(1-20 -a , b ,c , d and e).





a) Optic fiber cable[25] . b) Internet and.. wires. c)Abe device. d)Dancing fountain. e) Pictures of total Internal reflection.

Fig.(1-20- a , b ,c ,d and e)



1-4-d-Different kinds of Medical telescope[42]: Fig.(1-21 a, b and c)

The advantages of the endoscope :

1-The light propagate in straight lines even its curved tubes.

2- It keeps the strength of light at a constant level.

a)

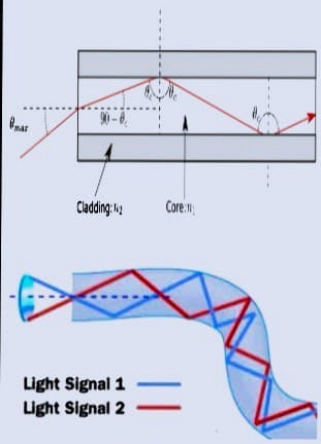
b)

c)

Fig.(1-21 a ,b and c)

meet.google.com is sharing your screen.
Stop sharing
Hide

1-6-c-Applications on the Critical angle and total internal reflection of light in industrial production as shown in Fig.(1-14 a , b ,c and d) .



a) Optic fiber.

b) Internet and.. wires.

c)Abe device.

d)Dancing fountain

1-6-d-Different kinds of Medical telescope: As shown in Fig.(1-14 a,b and c)

The advantages:

1-The light propagate in straight lines even its curved tubes.

2- It keeps the strength of light at a constant level.

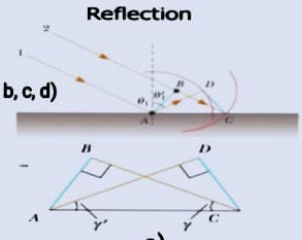
a)

b)

c)

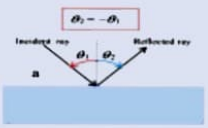
Fig.(1-14 a ,b and c)

1-7-Hygens's principle applied to Reflection and Refraction: as shown in Fig.(1-15 a, b, c, d)

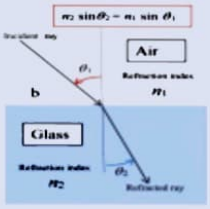


a)

b)



c)



d)

Fig.(1-15 a, b, c, d)

Hygins's principle depend on the Snell's wavelets (in his refraction law) and the idea of the congruent of the two triangle.

Reflection law: $\theta_1 = \theta_1{}'$

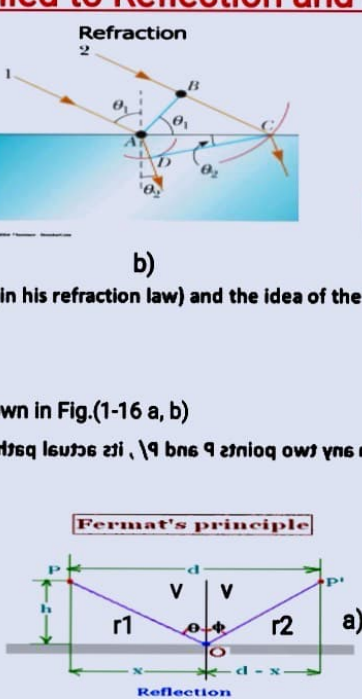
Refraction law: $n_1 sin\theta_1 = n_2 sin\theta_2$

1-8. Fermat's Principal: as shown in Fig.(1-16 a, b)

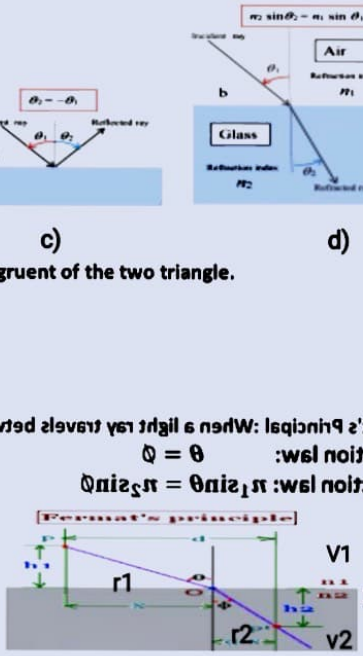
Fermat's Principal :When a light ray travels between any two points P and P\ , its actual path will be the one that requires the least time.

Reflection law: $\theta = \theta$

Refraction law: $n_1 sin\theta = n_2 sin\theta$



a)



b)

Fig.(1-16 a, b)